SCENARIO-BASED RESEARCH ASSESSMENT

Course :Artificial Intelligence and Expert Systems

Course Code:MLA0103

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1. Title of AI Application

PharmaVision: A Deep Learning-Based Multilingual Mobile System for Automated Expiry Validation and Counterfeit Medicine Detection

AI Domain: Healthcare Artificial Intelligence
Technologies: Deep Learning, Computer Vision, Optical Character Recognition, Machine Learning, Mobile AI, Cloud Computing

2. Domain Selection

Selected Domain: Healthcare

Healthcare is one of the most important application areas of Artificial Intelligence. AI is increasingly being used for medical diagnosis, drug discovery, patient monitoring, medical image analysis, and pharmaceutical supply-chain security.

This research focuses on PharmaVision, an AI-powered mobile application designed to help identify expired medicines and detect potentially counterfeit pharmaceutical products using a smartphone camera. The application combines Computer Vision, Deep Learning, Optical Character Recognition (OCR), and cloud-based data management to provide rapid medicine verification.

The problem is highly relevant because the World Health Organization identifies substandard and falsified medical products as a serious global public-health problem. Such products can fail to treat diseases, cause harm, contribute to antimicrobial resistance, and reduce public trust in healthcare systems

3. Introduction and Background

Medicines are essential for the prevention and treatment of diseases. However, two major problems affect pharmaceutical safety:

1. **Expired medicines**
2. **Counterfeit or falsified medicines**

Traditionally, users must manually check the expiry date printed on medicine packaging. This process can be difficult because expiry dates may be printed in small fonts, poor-quality packaging, or different formats. Similarly, identifying counterfeit medicines normally requires expert examination, laboratory testing, or verification through pharmaceutical authorities.

Recent developments in Artificial Intelligence have created opportunities to automate these processes. Computer Vision allows machines to analyse images, while OCR extracts printed text from photographs. Deep Learning models can learn visual differences between genuine and suspicious packaging.

PharmaVision proposes a smartphone-based solution in which the user captures an image of a medicine package. The system analyses the image, extracts important information such as the medicine name, batch number, manufacturing date, and expiry date, and evaluates visual and textual features to produce an intelligent verification result.

The application is designed as a multilingual mobile system to make the technology more accessible to users. The overall objective is to improve medicine safety through fast, accessible, and AI-assisted verification.

4. Problem Identification

The main problem addressed by PharmaVision is the difficulty of quickly identifying whether a medicine is:

- Expired
- Genuine
- Potentially counterfeit
- Incorrectly labelled
- Suspicious based on packaging or product information

Existing Problems

a. Manual expiry-date checking

Users usually need to locate and read small printed text manually. This can be difficult for:

- Elderly users
- Users with poor vision
- People unfamiliar with medicine packaging
- Medicines with unclear or damaged labels

b. Counterfeit medicine detection

Counterfeit medicines may imitate genuine packaging and may include:

- Fake logos
- Incorrect fonts
- Altered colours
- Fake batch numbers
- Incorrect packaging patterns
- Reused or duplicated product information

c. Lack of easily accessible verification tools

Many ordinary users do not have access to laboratory testing or pharmaceutical experts. A smartphone-based AI system can provide an initial screening mechanism.

d. Global pharmaceutical supply-chain complexity

Medicines may be manufactured, packaged, distributed, and sold across different countries. Complex supply chains and online commerce create additional opportunities for falsified products to reach consumers.

5. Need for an AI Solution

Traditional approaches have several limitations.

Traditional Method	**Limitation**
Manual reading of expiry date	Time-consuming and error-prone
Visual inspection	Depends heavily on human expertise
Laboratory testing	Expensive and not immediately available
Expert authentication	Not accessible to every user

Artificial Intelligence is required because the system must process several types of information simultaneously.

For example, the application may need to:

1. Capture an image.
2. Detect the medicine package.
3. Extract text using OCR.
4. Identify expiry-date patterns.

5. Compare packaging features.
6. Analyse the medicine against available records.
7. Generate an intelligent result.

AI can improve the traditional process by providing:

- Faster analysis
- Automated text extraction
- Consistent image analysis
- Reduced human effort
- Mobile accessibility
- Data-driven decision support

However, PharmaVision should be considered an AI-assisted screening and decision-support system, not a replacement for official pharmaceutical regulatory testing or laboratory authentication.

6. AI Technologies Used

6.1 Computer Vision

Computer Vision allows the system to analyse images captured using a smartphone camera.

It can be used to identify:

- Medicine packaging
- Labels
- Logos
- Barcodes
- Batch numbers
- Expiry-date regions
- Visual irregularities

The system can analyse the shape, colour, text layout, and visual characteristics of packaging.

6.2 Deep Learning

Deep Learning models can learn complex patterns from images.

A Convolutional Neural Network (CNN) or another image-classification architecture can be trained to classify medicine-package images into categories such as:

- Genuine
- Suspicious
- Counterfeit

A simplified CNN process is:

Input Image → Convolution → Feature Extraction → Classification → Result

Recent research has also demonstrated that deep neural networks can be used for counterfeit-product detection from smartphone images, showing the potential of computer-vision-based authentication systems.

6.3 Optical Character Recognition (OCR)

OCR converts text visible in an image into machine-readable text.

For example, an image may contain:

EXP: 08/2027

The OCR system extracts:

08/2027

The application can then compare this value with the current date.

OCR can also extract:

- Medicine name
- Batch number
- Manufacturing date
- Expiry date
- Manufacturer information

Google ML Kit provides Android text-recognition capabilities that can process an image and identify text from it.

6.4 Machine Learning Classification

After processing the image and extracted text, the system can classify the medicine.

Example:

Input:

- Medicine image
- OCR text

- Batch number
- Expiry date
- Packaging features

Processing:

- Image feature analysis
- Text verification
- Database comparison
- Classification model

Output:

- Genuine
- Expired
- Suspicious/Counterfeit Alert

6.5 Cloud Computing and Database Technology

Cloud services can store:

- User accounts
- Medicine records
- Scan history
- Inventory data
- Reports
- Suspicious product reports

For the PharmaVision application, Firebase can be used for services such as authentication, cloud storage, and database management.

6.6 Multilingual AI and Voice Support

A multilingual interface can make the system more accessible to users who may not be comfortable using English.

The application can provide results in languages such as:

- English
- Tamil
- Hindi

- Telugu

Voice alerts can also improve accessibility.

For example:

"Warning: This medicine may be expired."

7. Working Mechanism

The working process of PharmaVision can be explained through the following pipeline:

Step 1: User Authentication

The user logs into the application or creates an account.

Step 2: Medicine Scanning

The user opens the scanner and captures an image of the medicine package using the smartphone camera.

Step 3: Image Preprocessing

The image may be processed using:

- Cropping
- Resizing
- Noise reduction
- Contrast enhancement
- Image normalization

This improves the quality of the input image.

Step 4: Object and Region Detection

The system identifies relevant areas of the medicine package, such as:

- Product label
- Expiry-date region
- Batch number
- Logo
- Barcode

Step 5: OCR Processing

OCR extracts text from the image.

Example:

Captured Text:

PARACETAMOL 500 mg
BATCH: PCT2458
MFG: 08/2025
EXP: 07/2028

The system converts this visual information into structured data.

Step 6: Expiry-Date Validation

The system identifies the expiry date and compares it with the current date.

Example:

Current Date: July 2026
Expiry Date: July 2025

Result:

EXPIRED MEDICINE

If the expiry date is in the future:

Result:

NOT EXPIRED

Step 7: Counterfeit Detection

The deep-learning model analyses visual features such as:

- Packaging layout
- Logo
- Colour patterns
- Typography
- Label structure
- Product appearance

These features can be compared with known genuine product examples.

Step 8: Database Verification

The extracted information may be compared with available medicine records.

The system can examine:

- Product name
- Batch number
- Manufacturer
- Packaging information

Step 9: Result Generation

The application displays a final result.

Possible results include:

Genuine / Verified

The product information is consistent with the available data.

Expired Product

The expiry date has passed.

Counterfeit Alert

The product shows suspicious characteristics and requires further verification.

Step 10: Record Storage

The scan result can be stored in the user's history.

This allows the user or pharmacy staff to review previous scans.

Simplified System Workflow

User → Smartphone Camera → Image Preprocessing → OCR + Computer Vision → Deep Learning Model → Database Verification → AI Decision → Result and Alert

8. Data Sources and Processing

The system can use several types of data.

8.1 Image Data

Medicine-package images can be used for training and testing the computer-vision model.

Images may include:

- Genuine medicine packaging
- Counterfeit or suspicious packaging
- Different lighting conditions
- Different angles

- Damaged packaging
- Different backgrounds

8.2 Text Data

OCR extracts textual information such as:

- Medicine name
- Batch number
- Manufacturing date
- Expiry date
- Manufacturer name

8.3 Structured Medicine Data

A database may contain:

Field	**Example**
Medicine Name	Paracetamol
Batch Number	PCT2458
Manufacturer	Example Pharma
Manufacturing Date	08/2025
Expiry Date	07/2028

8.4 Data Processing Pipeline

Data Collection → Cleaning → Labelling → Image Preprocessing → Model Training → Testing → Deployment

The data must be carefully labelled because incorrect training data can negatively affect model performance.

The system should also be tested with different:

- Lighting conditions
- Camera qualities
- Packaging types
- Fonts

- Languages
- Image angles

9. Real-World Implementation

PharmaVision represents a practical application of AI in pharmaceutical safety.

Potential users include:

Consumers

Users can scan medicines before use or purchase.

Pharmacies

Pharmacists can use the application for preliminary verification and inventory monitoring.

Hospitals

Healthcare institutions can use automated scanning as part of medicine inventory processes.

Distributors

Distributors can use AI-assisted verification during supply-chain operations.

Auditors and Regulatory Authorities

AI systems can help identify suspicious products that require further investigation.

The global problem of falsified medical products has created the need for improved detection, reporting, and monitoring systems. WHO has also published guidance for identifying suspected substandard and falsified medical products.

A real-world example of the complexity of counterfeit medicine detection is the use of falsified batch numbers and repackaged products in counterfeit drug distribution. This demonstrates why relying on only one piece of information, such as a batch number, may not always be sufficient.

Therefore, PharmaVision's multi-modal approach—combining visual analysis, OCR, and database information—can provide stronger preliminary screening than relying only on manual inspection.

10. Impact and Benefits

10.1 Improved Safety

The system can help users identify expired or suspicious medicines before use.

10.2 Faster Verification

AI can analyse an image in a short period of time.

10.3 Reduced Manual Work

Users do not need to manually read every piece of information on a package.

10.4 Improved Accessibility

A mobile application makes AI-based medicine screening more accessible.

10.5 Support for Pharmacies

Pharmacists can use the system for inventory management and preliminary product screening.

10.6 Multilingual Accessibility

Language and voice support can improve usability for a wider population.

10.7 Digital Record Keeping

Scan history can help users maintain records of previously analysed medicines.

10.8 Potential Supply-Chain Support

The technology can potentially be extended to pharmaceutical distributors and healthcare organizations.

11. Challenges and Limitations

Despite its advantages, PharmaVision has several limitations.

11.1 Image Quality

Poor lighting, blur, glare, or low-resolution images can reduce the accuracy of both OCR and image classification.

11.2 Packaging Variations

The same medicine may have different packaging designs in different countries or production batches.

11.3 Counterfeit Adaptation

Counterfeiters may change packaging designs over time.

Therefore, AI models require continuous updating.

11.4 False Positives and False Negatives

The system may incorrectly identify a genuine product as suspicious or fail to detect a counterfeit product.

For this reason, an AI result should not be treated as absolute proof of authenticity.

11.5 Training Data Limitations

A model trained on a limited dataset may not perform well on unfamiliar medicine brands.

11.6 Privacy and Security

Medicine-related information and user data must be securely stored.

11.7 Internet Dependency

Cloud-based features may require internet connectivity.

11.8 Regulatory Requirements

Healthcare AI applications may need to comply with applicable medical-device, data-protection, and pharmaceutical regulations.

11.9 Ethical Concerns

Users may over-trust AI-generated results. Therefore, the application should clearly communicate that suspicious products require professional or regulatory verification.

12. Future Scope and Emerging Trends

The future development of PharmaVision can include several advanced features.

12.1 Advanced Deep Learning Models

More powerful models can improve counterfeit-product classification.

12.2 Multimodal AI

Future systems may combine:

- Image data
- OCR text
- Barcode information
- Batch information
- Supply-chain records

This can improve decision-making.

12.3 Blockchain-Based Pharmaceutical Traceability

Blockchain technology can be integrated to create a tamper-resistant record of a medicine's movement through the supply chain.

12.4 Edge AI

AI models can run directly on smartphones, allowing faster processing and reducing the need to upload sensitive images to the cloud.

12.5 Improved Multilingual AI

The system can support more Indian and international languages.

12.6 Voice-Based Interaction

Users could ask questions such as:

"Is this medicine expired?"

The system could provide a spoken answer.

12.7 Federated Learning

Future AI systems may learn from data across multiple organizations without directly sharing sensitive raw data.

12.8 Integration with Official Pharmaceutical Databases

Future versions could connect with verified pharmaceutical and regulatory databases for stronger authentication.

12.9 Generative AI Assistance

Generative AI could help explain scan results in simple language.

For example:

"This medicine has an expiry date of July 2025. Since the current date is July 2026, it should not be used without professional guidance."

However, generative AI should be carefully controlled in healthcare applications to prevent inaccurate medical advice.

13. Conclusion

PharmaVision demonstrates how Artificial Intelligence can be applied to a significant healthcare and pharmaceutical-safety problem.

The proposed system combines Deep Learning, Computer Vision, OCR, mobile computing, cloud services, and multilingual interaction to automate medicine analysis. The system can extract medicine information, validate expiry dates, analyse packaging characteristics, and provide an AI-assisted preliminary assessment.

The major advantage of this approach is that it brings AI-based analysis closer to ordinary users through a smartphone. It can reduce manual effort, improve accessibility, support pharmacies, and help identify medicines requiring further investigation.

However, AI should not replace official pharmaceutical authentication or laboratory testing. The most effective approach is to use AI as a rapid screening and decision-support tool combined with professional and regulatory verification.

Overall, AI-powered medicine verification represents an emerging healthcare application with significant potential for improving medicine safety, strengthening pharmaceutical monitoring, and reducing the risks associated with expired and falsified medical products.

14. References

You can use the following references in IEEE style:

[1] World Health Organization, “Substandard and falsified medical products,” WHO, 2026.

[2] World Health Organization, “Identifying suspected substandard and falsified medical products for health care professionals,” Technical Document, July 2025.

[3] World Health Organization, “Substandard and falsified medical products: Scope, scale and harm,” WHO, 2019.

[4] H. Garcia-Cotte, D. Mellouli, A. Rehman, L. Wang, and D. G. Stork, “Deep neural network-based detection of counterfeit products from smartphone images,” arXiv preprint arXiv:2410.05969, 2024.

[5] Google Developers, “Text Recognition | ML Kit,” Google for Developers.

[6] World Health Organization, “Digital transformation handbook for reporting substandard and falsified medical products,” WHO, 2026.

[7] Reuters, “How fake Ozempic batch numbers help criminal groups spread dangerous drugs,” 2024.